

hw3

1: Householder evals, det, and Singular vals

a) Evals

Let F be a householder matrix and $F = I - 2 \frac{vv^T}{v^T v}$

Then $Fv = v - 2 \frac{vv^T}{v^T v} v = v - 2 \frac{vv^T v}{v^T v} = v - 2v$ (The $v^T v$ in numerator and denom cancel)

So we have $Fv = -v$, using the eigenvalue equation: $Ax = \lambda x$, we deduce that one of the eigenvalues is -1 .

Now let x be such that the $\langle v, x \rangle = 0$ (x is orthogonal to v) and $\|x\| = 1$ so

$Fx = x - 2 \frac{vv^T}{v^T v} x = x - 2 \frac{vv^T x}{v^T v} = x$ (Fraction part gets eliminated since $v^T x$ is in the numerator and it equals 0)

So then we have $Fx = x$ and using the same eigenvalue equation we get the eigenvalue to be $+1$

Another proof is that the eigenvalue of all orthogonal symmetric matrices is 1 or -1 , we know that householder matrices are orthogonal so the eigenvalues must be 1

Can prove $F^T = F$ by $(I - 2 \frac{vv^T}{v^T v})^T = I - 2 \frac{vv^T}{v^T v} = F$

And then $F * F = (I - 2 \frac{vv^T}{v^T v})(I - 2 \frac{vv^T}{v^T v}) = I - 4 \frac{vv^T}{v^T v} + 4 \frac{vv^T vv^T}{v^T vv^T v}$ Cancel the $v^T v$ in top and bottom to get $= I - 4 \frac{vv^T}{v^T v} + 4 \frac{vv^T}{v^T v} = I$

So Then $F^T = F$ and $F^{-1} = F$

b) Det

We know that the determinant is the product of the eigenvalues. We know that v itself is an eigenvector with eigenvalue -1 . Similarly, x is an eigenvector if x is orthogonal to v . Since we can have $n-1$ such vectors (since n is the dimension and we already have 1 being v), the eigenvalue corresponding to x , 1 , has multiplicity $n-1$

So $\det(A) = -1 * 1^{n-1} = -1$

c) Singular Vals

Singular values are the square roots of the eigenvalues of the matrix $A^T A$. In this case since $F = F^T = F^{-1}$, $F^T F = I_n$ and the eigenvalues of the identity is just 1 (and $1 = \sqrt{1}$) so all singular values of F are 1

2) Householder and Norms

Problem 2. Consider a Householder matrix

$$F = I - 2 \frac{\vec{v}\vec{v}^T}{\vec{v}^T\vec{v}}, \quad \vec{v} \in R^n, \quad \vec{v} \neq \vec{0}.$$

(a) Show that if for a given $\vec{x} \in R^n$, $F\vec{x} = \vec{y}$, then $\|\vec{x}\|_2 = \|\vec{y}\|_2$.

(b) Show that if given $\vec{x}, \vec{y} \in R^n$ are such that $\vec{x} \neq \vec{y}$ and $\|\vec{x}\|_2 = \|\vec{y}\|_2$, then there is a Householder matrix F such that $F\vec{x} = \vec{y}$. **Hint:** Try $\vec{v} = \vec{x} - \vec{y}$.

a) norm x = norm y

We know that F is unitary so $\|F\vec{x}\| = \|\vec{x}\|$ so if we have $F\vec{x} = \vec{y}$, then $\|F\vec{x}\| = \|\vec{y}\|$ which means $\|\vec{x}\| = \|\vec{y}\|$

Proof for $\|F\vec{x}\| = \|\vec{x}\|$

$$\|F\vec{x}\|_2^2 = (F\vec{x})^T(F\vec{x}) = \vec{x}^T F^T F \vec{x} = \vec{x}^T \vec{x} = \|\vec{x}\|_2^2$$

b)

Know $F = I - 2 \frac{\vec{v}\vec{v}^T}{\vec{v}^T\vec{v}}$

So let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $\|\vec{x}\|_2 = \|\vec{y}\|_2$ but $\vec{x} \neq \vec{y}$ and set $\vec{v} = \vec{x} - \vec{y}$, then

$$F\vec{x} = \vec{x} - 2\vec{v} \frac{\vec{v}^T \vec{x}}{\vec{v}^T \vec{v}} = \vec{x} - 2(\vec{x} - \vec{y}) \frac{(\vec{x} - \vec{y})^T \vec{x}}{(\vec{x} - \vec{y})^T (\vec{x} - \vec{y})} = \vec{x} - 2(\vec{x} - \vec{y}) \frac{\vec{x}^T \vec{x} - \vec{y}^T \vec{x}}{\vec{x}^T \vec{x} - \vec{x}^T \vec{y} - \vec{y}^T \vec{x} + \vec{y}^T \vec{y}}$$

Since we have $\|\vec{x}\| = \|\vec{y}\|$, $\vec{x}^T \vec{x} = \vec{y}^T \vec{y}$ and since we are in $\mathbb{R}$, $\vec{x}^T \vec{y} = \vec{y}^T \vec{x}$ so

$$= \vec{x} - 2(\vec{x} - \vec{y}) \frac{\vec{x}^T \vec{x} - \vec{y}^T \vec{x}}{\vec{x}^T \vec{x} - \vec{x}^T \vec{y} - \vec{x}^T \vec{y} + \vec{y}^T \vec{x}}$$

$$= \vec{x} - 2(\vec{x} - \vec{y}) \frac{\vec{x}^T \vec{x} - \vec{y}^T \vec{x}}{2(\vec{x}^T \vec{x} - \vec{x}^T \vec{y})} \text{ Now we can cancel the numerator and denom (and the 2s)}$$

$$= \vec{x} - 2(\vec{x} - \vec{y}) \frac{1}{2}$$

$$= \vec{x} - \vec{x} + \vec{y}$$

$$= \vec{y}$$

So then now we have shown that $F\vec{x} = \vec{y}$ if F is made up of $\vec{v} = \vec{x} - \vec{y}$

3:

Code:

```
clear; clc;
function [Q, R] = clgs(A)

    n = size(A, 2);

    R = zeros(n);
    Q = zeros(size(A,1), n);
    for j = 1:n
        v = A(:, j);

        for i = 1:(j-1)
```

```

        R(i,j) = Q(:, i)' * A( :, j);
        v = v - R(i, j) * Q(:, i);

    end

    R(j,j) = norm(v);
    Q(:, j) = v / R(j,j);

end
end

function [Q, R] = mgs(A)

    n = size(A, 2);

    R = zeros(n);
    Q = zeros(size(A,1), n);

    V = zeros(size(A));

    for i = 1:n
        V(:, i) = A(:, i);
    end

    for i = 1:n

        R(i , i) = norm(V(:, i));
        Q( :, i) = V(:, i) / R(i,i);

        for j = (i+1):n
            R(i , j) = Q(:,i)' * V(:, j);
            V( :, j) = V( :, j) - R(i, j) * Q(:, i);
        end
    end
end

epsillons = [1e-4, 1e-6, 1e-8];
format long

function [Q, R] = house(A)
    [m,n] = size(A);

    R = zeros(n);
    Q = eye(m);

```

```

for k = 1:n
    x = A(k:m, k);

    e = zeros(length(x), 1);
    e(1) = 1;

    vk = sign(x(1)) * norm(x, 2) * e + x;
    vk = vk / norm(vk, 2);

    A(k:m, k:n) = A(k:m, k:n) - 2*vk * (vk' * A(k:m, k:n));
    Q( :, k:m) = Q( :, k:m) - 2* (Q( :, k:m) * vk) * vk';
end

R = A;

end

for e = epsillons
    A = [1, 1, 1; e, 0, 0; 0, e, 0; 0, 0, e];

    fprintf("Q and R for epsilon = %f using classical GSP", e)
    [Q, R] = clgs(A)
    fprintf("QR:")
    Q*R
    fprintf("Q^TQ:")
    Q' * Q

    fprintf("Q and R for epsilon = %f using modified GSP", e)
    [Q, R] = mgs(A)
    fprintf("QR:")
    Q * R
    fprintf("Q^TQ:")
    Q' * Q

    fprintf("Q and R for epsilon = %f using Householder", e)
    [Q, R] = house(A)
    fprintf("QR:")
    Q * R
    fprintf("Q^TQ:")
    Q' * Q

```

```

    fprintf("Q and R for epsilon = %f using built-in", e)
    [Q, R] = qr(A)
    fprintf("QR:")
    Q * R
    fprintf("Q^TQ:")
    Q' * Q
end

```

Output:

Q and R for epsilon = 0.000100 using classical GSP

Q =

0.999999995000000	0.000070710677866	0.000040824828876
0.000099999999500	-0.707106775883247	-0.408248287161463
0	0.707106782954315	-0.408248290364194
0	0	0.816496581608140

R =

1.000000005000000	0.999999995000000	0.999999995000000
0	0.000141421355884	0.000070710677866
0	0	0.000122474487037

QR:

ans =

1.000000000000000	1.000000000000000	1.000000000000000
0.000100000000000	-0.000000000000000	0
0	0.000100000000000	0
0	0	0.000100000000000

Q^TQ:

ans =

1.000000000000000	0.0000000000000277	0.0000000000000160
0.0000000000000277	1.000000000000000	-0.000000002264673
0.0000000000000160	-0.000000002264673	1.000000000000000

Q and R for epsilon = 0.000100 using modified GSP

Q =

0.999999995000000	0.000070710677866	0.000040824829036
0.000099999999500	-0.707106775883247	-0.408248288762828
0	0.707106782954315	-0.408248288762829

0	0	0.816496581608140
---	---	-------------------

R =

1.000000005000000	0.999999995000000	0.999999995000000
0	0.000141421355884	0.000070710677588
0	0	0.000122474487037

QR:

ans =

1.000000000000000	1.000000000000000	1.000000000000000
0.000100000000000	-0.000000000000000	0
0	0.000100000000000	0
0	0	0.000100000000000

Q^TQ:

ans =

1.000000000000000	0.0000000000000277	0.0000000000000160
0.0000000000000277	1.000000000000000	-0.000000000000000
0.0000000000000160	-0.000000000000000	1.000000000000000

Q and R for epsilon = 0.000100 using Householder

Q =

Columns 1 through 3

-0.999999995000000	0.000070710677588	0.000040824828876
-0.000099999999500	-0.707106775883246	-0.408248288762828
0	0.707106782954315	-0.408248288762828
0	0	0.816496581608140

Column 4

-0.000057735026823
0.577350268227375
0.577350268227375
0.577350268227375

R =

-1.000000005000000	-0.999999995000000	-0.999999995000000
0.000000000000000	0.000141421355884	0.000070710677588

0	0.0000000000000000	0.000122474487037
0	0	0

QR:

ans =

0.9999999999999999	0.9999999999999999	0.9999999999999999
0.0001000000000000	0.0000000000000000	0.0000000000000000
0.0000000000000000	0.0001000000000000	0.0000000000000000
0	0.0000000000000000	0.0001000000000000

Q^TQ :

ans =

Columns 1 through 3

0.9999999999999999	0.0000000000000000	0.0000000000000000
0.0000000000000000	1.0000000000000000	-0.0000000000000000
0.0000000000000000	-0.0000000000000000	1.0000000000000000
-0.0000000000000000	0.0000000000000000	0.0000000000000000

Column 4

-0.0000000000000000
0.0000000000000000
0.0000000000000000
1.0000000000000000

Q and R for epsilon = 0.000100 using built-in

Q =

Columns 1 through 3

-0.999999995000000	0.000070710677588	0.000040824828876
-0.000099999999500	-0.707106775883247	-0.408248288762829
0	0.707106782954315	-0.408248288762829
0	0	0.816496581608140

Column 4

-0.000057735026823
0.577350268227375
0.577350268227375
0.577350268227375

R =

-1.000000005000000	-0.999999995000000	-0.999999995000000
0	0.000141421355884	0.000070710677588
0	0	0.000122474487037
0	0	0

QR:

ans =

1.000000000000000	1.000000000000000	1.000000000000000
0.000100000000000	-0.000000000000000	-0.000000000000000
0	0.000100000000000	0
0	0	0.000100000000000

Q^TQ:

ans =

Columns 1 through 3

1.000000000000000	0	0
0	1.000000000000000	0
0	0	1.000000000000000
-0.000000000000000	-0.000000000000000	-0.000000000000000

Column 4

-0.000000000000000
-0.000000000000000
-0.000000000000000
1.000000000000000

Q and R for epsilon = 0.000001 using classical GSP

Q =

0.999999999999500	0.000000707169643	0.000000408248287
0.000001000000000	-0.707106781186017	-0.408211996414818
0	0.707106781186724	-0.408284583437063
0	0	0.816496579852289

R =

1.000000000000500	0.999999999999500	0.999999999999500
0	0.000001414213562	0.000000707169643
0	0	0.000001224744873

QR:

ans =

1.0000000000000000	1.0000000000000000	1.0000000000000000
0.0000010000000000	0	0
0	0.0000010000000000	0
0	0	0.0000010000000000

Q^TQ :

ans =

1.0000000000000000	0.0000000000062862	0.0000000000036290
0.0000000000062862	1.0000000000000000	-0.000051326775655
0.0000000000036290	-0.000051326775655	1.0000000000000000

Q and R for epsilon = 0.000001 using modified GSP

Q =

0.999999999999500	0.000000707169643	0.000000408284584
0.0000010000000000	-0.707106781186017	-0.408248290463693
0	0.707106781186724	-0.408248290463693
0	0	0.816496580927794

R =

1.0000000000000500	0.999999999999500	0.999999999999500
0	0.000001414213562	0.000000707106781
0	0	0.000001224744871

QR:

ans =

1.0000000000000000	1.0000000000000000	1.0000000000000000
0.0000010000000000	0	0
0	0.0000010000000000	0
0	0	0.0000010000000000

Q^TQ :

ans =

1.0000000000000000	0.0000000000062862	0.0000000000036294
0.0000000000062862	1.0000000000000000	-0.0000000000000000
0.0000000000036294	-0.0000000000000000	1.0000000000000000

Q and R for epsilon = 0.000001 using Householder

Q =

Columns 1 through 3

-0.999999999999500	0.000000707106781	0.000000408248290
-0.000001000000000	-0.707106781186017	-0.408248290463693
0	0.707106781186724	-0.408248290463693
0	0	0.816496580927794

Column 4

-0.000000577350269
0.577350269189529
0.577350269189529
0.577350269189530

R =

-1.0000000000000500	-0.999999999999500	-0.999999999999500
0	0.000001414213562	0.000000707106781
0	0	0.000001224744871
0	0	0.000000000000000

QR:

ans =

1.000000000000000	1.000000000000000	1.000000000000000
0.000001000000000	-0.000000000000000	-0.000000000000000
0	0.000001000000000	0.000000000000000
0	0	0.000001000000000

Q^TQ:

ans =

Columns 1 through 3

1.000000000000000	-0.000000000000000	-0.000000000000000
-0.000000000000000	1.000000000000000	0.000000000000000
-0.000000000000000	0.000000000000000	1.000000000000000
0	-0.000000000000000	0.000000000000000

Column 4

0

```
-0.0000000000000000
0.0000000000000000
1.0000000000000000
```

Q and R for epsilon = 0.000001 using built-in

Q =

Columns 1 through 3

```
-0.999999999999500    0.000000707106781    0.000000408248290
-0.000001000000000  -0.707106781186017  -0.408248290463693
                        0    0.707106781186724  -0.408248290463693
                        0                        0    0.816496580927794
```

Column 4

```
-0.000000577350269
0.577350269189530
0.577350269189529
0.577350269189529
```

R =

```
-1.000000000000500  -0.999999999999500  -0.999999999999500
                        0    0.000001414213562    0.000000707106781
                        0                        0    0.000001224744871
                        0                        0                        0
```

QR:

ans =

```
1.000000000000000    1.000000000000000    1.000000000000000
0.000001000000000    0.000000000000000    0.000000000000000
                        0    0.000001000000000                        0
                        0                        0    0.000001000000000
```

Q^TQ:

ans =

Columns 1 through 3

```
1.000000000000000    0                        0
                        0    1.000000000000000  -0.000000000000000
                        0  -0.000000000000000    1.000000000000000
                        0    0.000000000000000    0.000000000000000
```

Column 4

```
      0
0.0000000000000000
0.0000000000000000
1.0000000000000000
```

Q and R for epsilon = 0.000000 using classical GSP

Q =

```
1.0000000000000000      0      0
0.0000000010000000 -0.707106781186548 -0.707106781186548
      0      0.707106781186548      0
      0      0      0.707106781186548
```

R =

```
1.0000000000000000      1.0000000000000000      1.0000000000000000
      0      0.0000000014142136      0
      0      0      0.0000000014142136
```

QR:

ans =

```
1.0000000000000000      1.0000000000000000      1.0000000000000000
0.0000000010000000      0      0
      0      0.0000000010000000      0
      0      0      0.0000000010000000
```

Q^TQ:

ans =

```
1.0000000000000000 -0.000000007071068 -0.000000007071068
-0.000000007071068 1.0000000000000000 0.5000000000000000
-0.000000007071068 0.5000000000000000 1.0000000000000000
```

Q and R for epsilon = 0.000000 using modified GSP

Q =

```
1.0000000000000000      0      0
0.0000000010000000 -0.707106781186548 -0.408248290463863
      0      0.707106781186548 -0.408248290463863
      0      0      0.816496580927726
```

R =

1.0000000000000000	1.0000000000000000	1.0000000000000000
0	0.000000014142136	0.000000007071068
0	0	0.000000012247449

QR:

ans =

1.0000000000000000	1.0000000000000000	1.0000000000000000
0.0000000100000000	0	0
0	0.0000000100000000	0
0	0	0.0000000100000000

Q^TQ :

ans =

1.0000000000000000	-0.000000007071068	-0.000000004082483
-0.000000007071068	1.0000000000000000	-0.0000000000000000
-0.000000004082483	-0.0000000000000000	1.0000000000000000

Q and R for epsilon = 0.000000 using Householder

Q =

Columns 1 through 3

-1.0000000000000000	0.000000007071068	0.000000004082483
-0.0000000100000000	-0.707106781186547	-0.408248290463863
0	0.707106781186548	-0.408248290463863
0	0	0.816496580927726

Column 4

-0.000000005773503
0.577350269189626
0.577350269189626
0.577350269189626

R =

-1.0000000000000000	-1.0000000000000000	-1.0000000000000000
0	0.000000014142136	0.000000007071068
0	-0.0000000000000000	0.000000012247449
0	0	-0.0000000000000000

QR:
ans =

1.0000000000000000	1.0000000000000000	1.0000000000000000
0.0000000100000000	-0.0000000000000000	-0.0000000000000000
0	0.0000000100000000	-0.0000000000000000
0	-0.0000000000000000	0.0000000100000000

Q^TQ:
ans =

Columns 1 through 3

1.0000000000000000	0	-0.0000000000000000
0	1.0000000000000000	-0.0000000000000000
-0.0000000000000000	-0.0000000000000000	1.0000000000000000
0.0000000000000000	0	-0.0000000000000000

Column 4

0.0000000000000000
0
-0.0000000000000000
1.0000000000000000

Q and R for epsilon = 0.000000 using built-in
Q =

Columns 1 through 3

-1.0000000000000000	0.000000007071068	0.000000004082483
-0.0000000100000000	-0.707106781186548	-0.408248290463863
0	0.707106781186547	-0.408248290463863
0	0	0.816496580927726

Column 4

-0.000000005773503
0.577350269189626
0.577350269189626
0.577350269189626

R =

```

-1.0000000000000000 -1.0000000000000000 -1.0000000000000000
      0      0.000000014142136      0.000000007071068
      0      0      0.000000012247449
      0      0      0

```

QR:

ans =

```

1.0000000000000000 1.0000000000000000 1.0000000000000000
0.0000000100000000 -0.0000000000000000 -0.0000000000000000
      0      0.0000000100000000 0.0000000000000000
      0      0      0.0000000100000000

```

Q^TQ:

ans =

Columns 1 through 3

```

1.0000000000000000      0      0
      0      1.0000000000000000 0.0000000000000000
      0      0.0000000000000000 1.0000000000000000
      0 -0.0000000000000000 -0.0000000000000000

```

Column 4

```

      0
-0.0000000000000000
-0.0000000000000000
1.0000000000000000

```

4:

Problem 4. (Necessary condition for a minimum using calculus.) Let $A \in R^{m \times n}$ and $\vec{b} \in R^m$ be given. If the function $f(\vec{x}) = \|\vec{b} - A\vec{x}\|_2^2$, $\vec{x} \in R^n$, has a minimum at $\vec{x}' \in R^n$, then $\partial f(\vec{x}')/\partial x_i = 0$, $i = 1, \dots, n$. Compute these equations and see if they agree with those derived in class using a different method.

Problem 5. Assume that $A \in R^{m \times n}$, $\vec{b} \in R^m$ are given. Let $\vec{x}' \in R^n$ be such that $A^T A \vec{x}' = A^T \vec{b}$. Show that

$$f(x) = \|b - Ax\|_2^2 = (b - Ax)^T(b - Ax) = b^T b - b^T Ax - x^T A^T b + x^T A^T Ax \text{ and}$$

$b^T Ax$ is a scalar because b^T is $1 \times m$, A is $m \times n$ and x is $n \times 1$

So then $x^T A^T b = (b^T Ax)^T$ and because transpose of the scalar is the same, we can combine

We combine $-b^T Ax - x^T A^T b = -2(x^T A^T b)$

$f(x) = b^T b - 2(x^T A^T b) + x^T A^T A x$ Now we can differentiate

$f'(x) = -2(A^T b) + 2A^T A x$ because $b^T b$ gets eliminated and the second term is linear while the third is quadratic

Now evaluating at $f'(x') = 0$

$$0 = -2A^T b + 2A^T A x$$

$$2A^T b = 2A^T A x$$

$$A^T b = A^T A x$$

This is the equation that we get in class so they agree

5:

Problem 5. Assume that $A \in R^{m \times n}$, $\vec{b} \in R^m$ are given. Let $\vec{x}' \in R^n$ be such that $A^T A \vec{x}' = A^T \vec{b}$. Show that

$$\|\vec{b} - A\vec{x}\|_2^2 = \|\vec{b} - A\vec{x}'\|_2^2 + \|A(\vec{x} - \vec{x}')\|_2^2 \quad \text{for every } \vec{x} \in R^n.$$

Conclude from this that if $\vec{x}' \in R^n$ is a solution of the system $A^T A \vec{x}' = A^T \vec{b}$, then $\vec{x}'$ minimizes $\|\vec{b} - A\vec{x}\|_2$ with respect to $\vec{x} \in R^n$.

$$\|b - Ax\|_2^2 = \|b - Ax + Ax' - Ax'\|_2^2 = \|b - Ax' + A(x' - x)\|_2^2$$

From here we can use a trick where

$$\|x + y\|^2 = (x + y)^T (x + y) = x^T x + x^T y + y^T x + y^T y \text{ and since we are in the reals, } x^T y = y^T x$$

$$\text{so } \|x\|^2 + 2x^T y + \|y\|^2$$

$$= \|b - Ax'\|_2^2 + \|A(x' - x)\|_2^2 + 2(b - Ax')^T (A(x' - x))$$

The non norm part is as follows:

$$2(b - Ax')^T (Ax' - Ax) = 2(b^T Ax' - b^T Ax - x'^T A^T Ax' + x'^T A^T Ax)$$

$$= 2(b^T Ax' - b^T Ax - x'^T A^T b + x'^T A^T Ax) \text{ bc } A^T Ax' = A^T b$$

$$(A^T Ax')^T = (A^T b)^T \text{ so } x'^T A^T A = b^T A$$

$$= 2(b^T Ax' - b^T Ax - x'^T A^T b + b^T Ax) \text{ From above}$$

$$= 2(b^T Ax' - x'^T A^T b) \text{ From above}$$

$$= 2(b^T Ax' - (b^T Ax')^T)$$

$b^T Ax'$ is a scalar because b^T is $1 \times m$, A is $m \times n$ and x' is $n \times 1$

And the transpose of a scalar is the same scalar so this is something minus itself which is 0

$$= 2 * 0 = 0$$

So the stuff passed the norm is 0

$$= \|b - Ax'\|_2^2 + \|A(x' - x)\|_2^2 \text{ and we can multiply the norm by -1 without changing its val so}$$

$$= \|b - Ax'\|_2^2 + \|A(x - x')\|_2^2$$

$$\text{So } \|b - Ax\|_2^2 = \|b - Ax'\|_2^2 + \|A(x - x')\|_2^2$$

Now let x' be a solution, then we know that $\|A(x' - x)\| \geq 0$ by the defn of norms. So this means that

$$\|b - Ax\|_2^2 \geq \|b - Ax'\|_2^2$$

So x' minimizes $\|b - Ax\|_2^2$

6)

For any x , $Ax = a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4$ and since $a_2 = \alpha a_1 + \beta a_3$

$$= a_1x_1 + (\alpha a_1 + \beta a_3)x_2 + a_3x_3 + a_4x_4$$

$$= a_1x_1 + \alpha a_1x_2 + \beta a_3x_2 + a_3x_3 + a_4x_4$$

$$= (x_1 + \alpha x_2)a_1 + (\beta x_2 + x_3)a_3 + a_4x_4$$

So if we let $y_1 = x_1 + \alpha x_2$, $y_2 = \beta x_2 + x_3$ and $y_3 = x_4$ then we have

$$= y_1a_1 + y_2a_3 + y_3a_4 \text{ then if we call } y = [y_1 y_2 y_3] \text{ and } B = [a_1 | a_3 | a_4] \text{ we get } Ax = By$$

This means that $\|b - Ax\|_2 = \|b - By\|_2$

So then if we rearrange $y_1 = x_1 + \alpha x_2$, $y_2 = \beta x_2 + x_3$ and $y_3 = x_4$ to be in terms of x , we get

$$x_1 = y_1 - \alpha x_2$$

$$x_3 = y_2 - \beta x_2$$

$$x_4 = y_3$$

And x_2 is free

So $x = [y_1 - t\alpha \quad t \quad y_2 - t\beta \quad y_3]^T$ where $t \in \mathbb{R}$ (here t is just x_2) (also I don't know how to do vertical arrays in LaTeX)

So we are done